

largest battery manufacturers based there. China controls the lithium supply chain, possessing 61% of the global lithium refining capacity. The country has made significant progress in mining rare materials, training engineers, and establishing large factories to meet the rising demand for batteries. Chinese companies are also leading the race in developing SIB manufacturing technology. The Chinese 14th Five-Year Plan for Scientific and Technological Innovation in the Energy Sector, which extends until 2025, emphasises research on SIBs to promote the use of renewable energy. CATL, the world's largest battery manufacturer, is at the forefront and plans to mass-produce and supply SIBs in 2023 at a lower cost compared to LIBs. In the next two years, China is expected to hold nearly 95% of the world's SIB manufacturing capacity. However, China faces challenges as it cannot rely on importing soda ash from the US due to sanctions. Soda ash is a primary industrial source of sodium, and the US accounts for over 90% of the world's mined reserves. As a result, China is compelled to produce synthetic soda ash, which poses a serious risk of hazardous water pollution. Recent incidents, such as the collapse of a pile of alkali slag contaminating a major



The future of SIBs also depends on future design of electric vehicles (Credit: <https://auto.economictimes.indiatimes.com>)

river in China, highlight the environmental concerns associated with synthetic soda ash production.

Future challenges

While SIBs offer a promising alternative to the widely used LIBs, their development is still in its nascent stage compared to LIBs. The increasing demand for lithium, cobalt, graphite, nickel, and manganese presents significant environmental and health risks. Given the scarcity of rare earth metals and the heavy reliance on China for the global supply chain, it is crucial to establish effective alternatives. SIBs, with their abundance of sodium, faster charging capabilities, wider temperature range, non-flammability, and lower cost, present a viable solution. However, there are several challenges to address. These include limitations in flexibility, as SIBs cannot be shaped into various forms like prismatic or cylindrical; lower energy density and storage capacity compared to LIBs; a typical cycle life of 5,000 cycles, significantly lower than the 8,000-10,000 cycles of LIBs; the lack of a well-established supply chain for raw materials; and the need for commercial-scale production. SIBs face limitations in applications such as deep-

sea exploration, aerospace technology, and operation of military equipment at very low temperatures due to sluggish electrochemical reactions and unstable interfacial reactions caused by extremely low operating temperatures.

Overcoming these challenges requires continuous research and development efforts, with a focus on improving electrolytes for

high charge and discharge rates over a wide temperature range, extending cycle life and shelf life, and gaining a deeper understanding of ion transport relationships in sodium intercalation electrodes.

SIBs have shown suitability for applications such as electric two-wheelers, three-wheelers, and low-speed short-range EVs. However, the larger size of sodium batteries compared to lithium batteries with the same electric charge poses a challenge for electric cars, which have limited space. Battery energy storage systems and power backup in electric utilities are other applications where energy density demands are less stringent. Intensive research is required to achieve specific energies approaching 200Wh/kg, a deep discharge cycle life exceeding 10,000 cycles, volumetric energy density greater than 20kWh per cubic metre, and a lifetime of over 10 years for high-voltage grid-scale applications. Therefore, further research and development are necessary to overcome these challenges and establish SIBs as a backbone of the rechargeable battery industry. Optimisation of performance and durability, improvements in materials and electrode designs, and enhancements in energy density and cycle life will determine when SIBs become commercially feasible and widely adopted. **EFY**



China leads the world's LIB and is ahead in the SIB battery market as well (Credit: <https://www.mobilityindia.com>)